



Separation of ethanol from *n*-decane-ethanol mixtures using imidazolium based ionic liquids



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ABSTRACT

Great features of ionic liquids (ILs) lead to consider them as new generation of solvents in liquid-liquid extraction process. In present work, 1-ethyl-3-methylimidazolium thiocyanate ([Emim][SCN]) and 1-ethyl-3-methylimidazolium trifluoromethanesulfonate ([Emim][Otf]) were utilized as solvents for separation of ethanol from *n*-decane-ethanol mixtures. The experiments were conducted to obtain liquid-liquid equilibria (LLE) data of *n*-decane/ethanol/[Emim][SCN] and *n*-decane/ethanol/[Emim][Otf] systems at two different temperatures ($T = 298.15$ K and $T = 313.15$ K) and the pressure of 101.3 kPa. The solute distribution coefficients (β) and selectivities (S) were calculated to investigate the efficiencies of [Emim][SCN] and [Emim][Otf] as solvents. Moreover, the temperature dependencies of β and S were investigated in this study. The non-random two liquid (NRTL) model was used to correlate the recorded experimental data of studied systems.

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1. Introduction

Today, releasing of volatile organic solvents from chemical industries to the atmosphere is one of the important environmental concerns [1]. Ionic liquids (ILs) as new classes of solvents are received attention due to unique properties in separation processes. Their properties such as: non-flammability, good capacity to solve organic/inorganic compounds, high thermal/chemical stability, low vapour pressure at room temperature, liquid state in wide temperature range, high heat capacity and recyclability are the features that have been made ILs as suitable alternative for conventional volatile organic solvents [2–4].

Liquid-liquid extraction technique besides other processes are employed to separate liquid mixtures into the pure components. This method can be performed in a simple equipment, under mild operating conditions, lower pressure and temperature, and requires less energy in comparison to other separation techniques [5–8]. These advantages over conventional separation methods such as distillation make it a suitable technique for purification of liquid mixtures. However, utilizing proper solvent plays a key role in separation efficiency and economical operation of liquid-liquid extraction process. Experimental liquid-liquid equilibria (LLE) data are required to find out the performance of a solvent

in separation process. Moreover, LLE data are required to obtain essential information for process design. Extraction of oxygenates, produced in side reactions in Fischer-Tropsch process, from hydrocarbon stream is mandatory as both oxygenates and alkanes are coexisting as process products [9,10]. As hydrocarbon stream includes variety of components and difficulty in stream modelling, normal decane (C_{10}) would be considered as the basis for modelling the hydrocarbon stream in this study. Although, a number of investigation have been conducted using ILs to separate azeotropic mixtures of alcohols and light alkanes ($<C_8$) [11–23], there are few studies investigate ILs as solvent for extraction of alcohols from heavier alkanes. Letcher et al. [24] investigated the effectiveness of using 1-methyl-3-octylimidazolium chloride for extraction of ethanol from dodecane and hexadecane. Ram et al. [25] used three alkylsulfate-based ILs to separate ethanol from its mixture with dodecane. They concluded that 1,3-dimethylimidazolium methylsulfate is more efficient in comparison to other examined ILs. Cai and Xiao [26] used dialkylphosphate-based ILs (1,3-dimethylimidazolium dimethylphosphate, 1-ethyl-3-methylimidazolium diethylphosphate, 1-butyl-3-methylimidazolium dibutylphosphate) as solvents to extract methanol from its mixtures with hexane and heptane. Their results showed that ILs with shorter cation chains length have more capability to separate methanol from hexane and heptane (e.g., 1,3-dimethylimidazolium dimethylphosphate). Xu et al. [27] examined the chain length effect of alcohols (ethanol, 1-propanol, 2-propanol) on the LLE in alcohol/heptane/IL systems. They concluded that ethanol can be separated

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more easily in comparison to 1-propanol and 2-propanol. Also, the feasibility of using four different imidazolium based ILs to extract butanol from its mixtures with heptane was investigated by Wen et al. [28]. They analysed the effect of the structure of anions on the extraction parameters.

In this work, 1-ethyl-3-methylimidazolium thiocyanate ([Emim][SCN]) and 1-ethyl-3-methylimidazolium trifluoromethanesulfonate ([Emim][OTf]) were studied as solvents to extract ethanol from its mixtures with *n*-decane and the experimental ternary LLE data for *n*-decane/ethanol/[Emim][SCN] and *n*-decane/ethanol/[Emim][OTf] systems were recorded at two temperatures: $T = 298.15$ K and $T = 313.15$ K. The experiments were carried out under the pressure of 101.3 kPa. Liquid-liquid extraction parameters include: selectivity and solute distribution ratio were obtained to investigate the feasibility of using [Emim][SCN] and [Emim][OTf] as extractive solvents. Additionally, NRTL model was used to correlate obtained LLE data.

2. Experimental

2.1. Chemicals

The ionic liquids: [Emim][OTf] and [Emim][SCN] used in the present research were purchased from Sigma-Aldrich, with minimum mass fraction purity 0.98 and 0.95, respectively. Ethanol and *n*-decane were supplied by Merck and Samchun companies with minimum mass fraction purities 0.999 and 0.980, respectively. The purities of ethanol and *n*-decane were assessed by gas chromatography (GC). Further information about chemicals used in this study are presented in Table 1.

2.2. Experimental procedure

In order to measure experimental equilibrium composition between two phases, 10 g of ternary mixture with specified

Table 1

Supplier, purity, water content by mass (w_w), and physical properties of studied materials at 298.15 K.

Chemical	Supplier	CAS number	Molecular formula	Molar mass/g·mol ⁻¹	Mass fraction purity	$w_w/10^{-6}$	Analysis method
Ethanol	Merck	64-17-5	C ₂ H ₆ O	46.07	≥0.999	<500 ^a	GC
<i>n</i> -decane	Samchun	124-18-5	C ₁₀ H ₂₂	142.28	≥0.980	–	GC
[Emim][SCN]	Sigma-Aldrich	331717-63-6	C ₇ H ₁₁ N ₃ S	169.25	≥0.950	<1000 ^a	HPLC ^a
[Emim][OTf]	Sigma-Aldrich	145022-44-2	C ₇ H ₁₁ F ₃ N ₂ O ₃ S	260.23	≥0.980	<500 ^a	H-NMR ^a

^a Analysis by supplier.

Table 2

Experimental LLE data for *n*-decane/ethanol/IL system in mole fraction, solute distribution ratio (β), and selectivity (S) at $T = 298.15$ – 313.15 K and pressure of 101.3 kPa.^a

T/K	Upper phase		Lower phase		β	S
	x_1^I	x_2^I	x_1^{II}	x_2^{II}		
<i>n</i> -Decane (1) + Ethanol (2) + [Emim][SCN] (3)						
298.15	1.000	0.000	0.001	0.000		
	0.992	0.008	0.001	0.131	16.78	13520.17
	0.973	0.027	0.002	0.311	11.37	5647.38
	0.955	0.045	0.002	0.426	9.55	3873.76
	0.942	0.058	0.003	0.504	8.69	2445.88
	0.925	0.075	0.004	0.594	7.90	1714.40
	0.890	0.110	0.006	0.677	6.17	916.46
	0.837	0.163	0.014	0.780	4.77	275.69
	0.754	0.246	0.036	0.854	3.47	73.24
313.15	0.992	0.008	0.001	0.104	12.27	8591.32
	0.970	0.030	0.002	0.288	9.46	4914.20
	0.956	0.044	0.003	0.379	8.60	3224.32
	0.939	0.061	0.003	0.469	7.70	2376.70
	0.926	0.074	0.004	0.560	8.54	1689.84
	0.886	0.114	0.004	0.713	6.25	1262.94
	0.810	0.190	0.015	0.799	4.20	232.72
	0.752	0.248	0.027	0.834	3.37	92.63
<i>n</i> -Decane (1) + Ethanol (2) + [Emim][Otf] (3)						
298.15	1.000	0.000	0.003	0.000		
	0.994	0.006	0.005	0.213	37.58	7987.32
	0.974	0.026	0.005	0.336	12.74	2333.97
	0.942	0.058	0.006	0.489	8.38	1365.07
	0.903	0.097	0.007	0.631	6.50	801.41
	0.865	0.135	0.012	0.735	5.45	403.76
	0.824	0.176	0.017	0.794	4.52	215.42
	0.731	0.269	0.036	0.896	3.33	68.47
313.15	0.990	0.010	0.004	0.198	19.38	4562.46
	0.973	0.027	0.004	0.298	10.89	2396.86
	0.954	0.046	0.005	0.447	9.73	2009.74
	0.943	0.057	0.005	0.475	8.36	1592.96
	0.879	0.121	0.008	0.619	5.11	574.29
	0.857	0.143	0.011	0.714	5.01	405.76
	0.826	0.174	0.014	0.789	4.53	269.39
	0.701	0.299	0.027	0.871	2.91	75.38

^a Standard uncertainties u are $u(x_i) = \pm 0.005$, $u(T) = 0.2$ K, $u(p) = 1$ kPa.

composition was placed in a glass with 20 mL volume containing a magnetic stirrer. The composition of mixture was given so that the mixture lies in immiscible region of system. An analytical balance with precision of ± 0.00001 g was used to weight mixtures. Two immiscible phases were stirred for 4 h to ensure complete combination of mixtures. The samples were left to settle down for at least 15 h at desired temperatures ($T = 298.15$ K and $T = 313.15$ K) to be entirely separated. Gas chromatography (YL-6100 GC, South Korea) was utilized to determine ethanol

and *n*-decane compositions. GC was equipped with a 60 m length, 1 μ m thickness, 0.32 mm in diameter Suprawax 280 capillary column and a flame ionization detector (FID), using helium as carrier gas. The samples were withdrawn from both upper and lower layers and injected to GC column using a syringe to analyze composition of each phase. Due to very low vapour pressure of ILs, their compositions were obtained after determining of *n*-decane and ethanol composition through GC analysis.

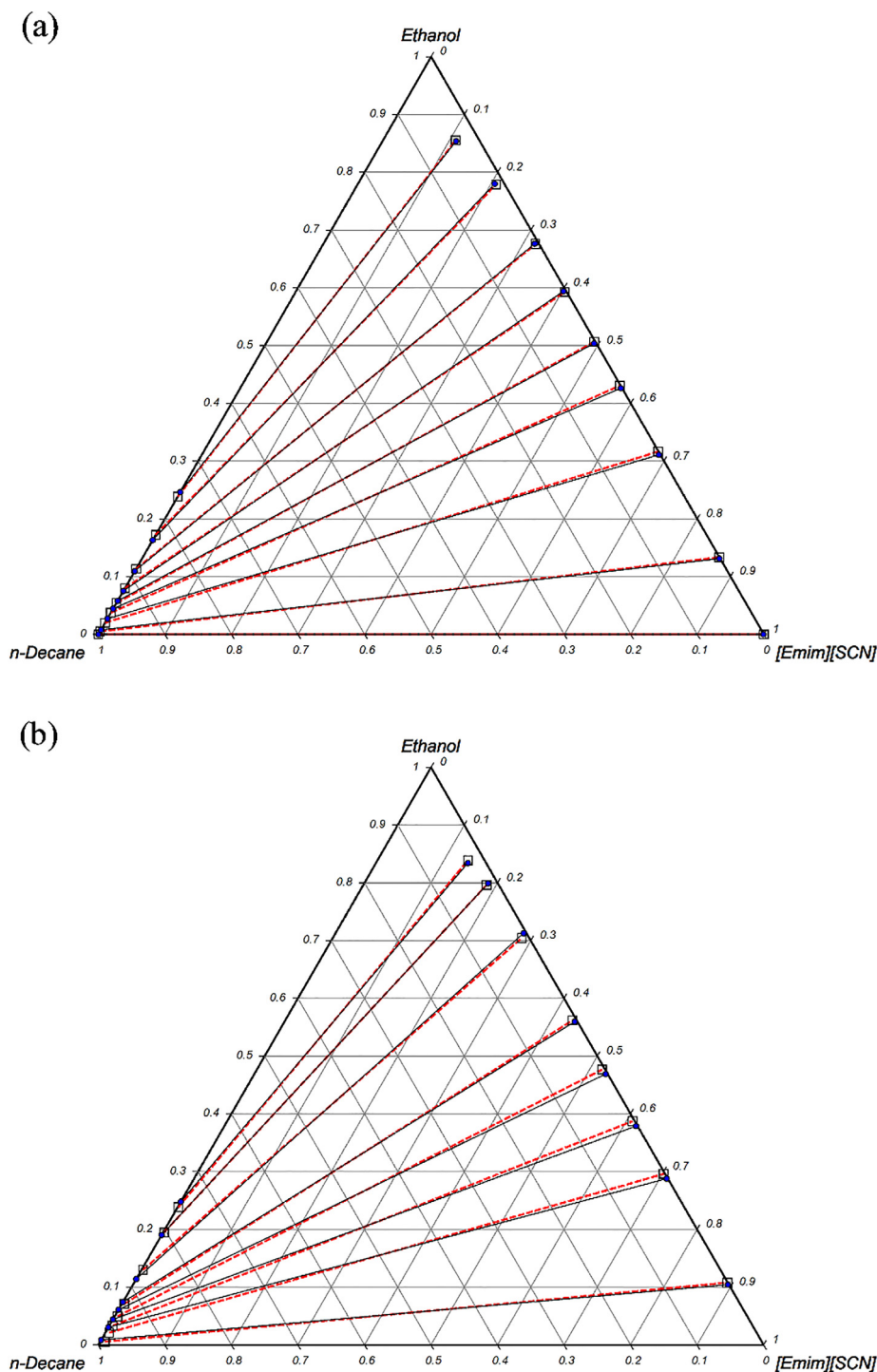


Fig. 1. Tie-lines of *n*-decane/ethanol/[Emim][SCN] ternary systems (a) at $T = 298.15$ K; (b) $T = 313.15$ K, and $p = 101.3$ kPa. (— with circles: experimental data; - - - with squares: NRTL model.)

3. Results and discussion

3.1. Liquid-liquid equilibria

The experimental LLE data for *n*-decane/ethanol/[Emim][SCN] and *n*-decane/ethanol/[Emim][Otf] systems at two different temperatures: $T = 298.15$ K and $T = 313.15$ K, under the pressure of 101.3 kPa are presented in Table 2. According to this Table *n*-decane is less soluble in [Emim][SCN], in comparison to [Emim][Otf].

The triangular phase diagram containing experimental tie lines of investigated ternary mixtures are illustrated in Figs. 1 and 2. The positive slope of tie lines indicate that ethanol has more tendency to be solved in the studied ILs than *n*-decane due to polar structure of ethanol and π - π interactions between the cation of IL and ethanol. As shown in Figs. 1 and 2, the slope of tie lines and two phase region of both *n*-decane/ethanol/[Emim][SCN] and *n*-decane/ethanol/[Emim][Otf] systems are slightly decreased by increasing temperature.

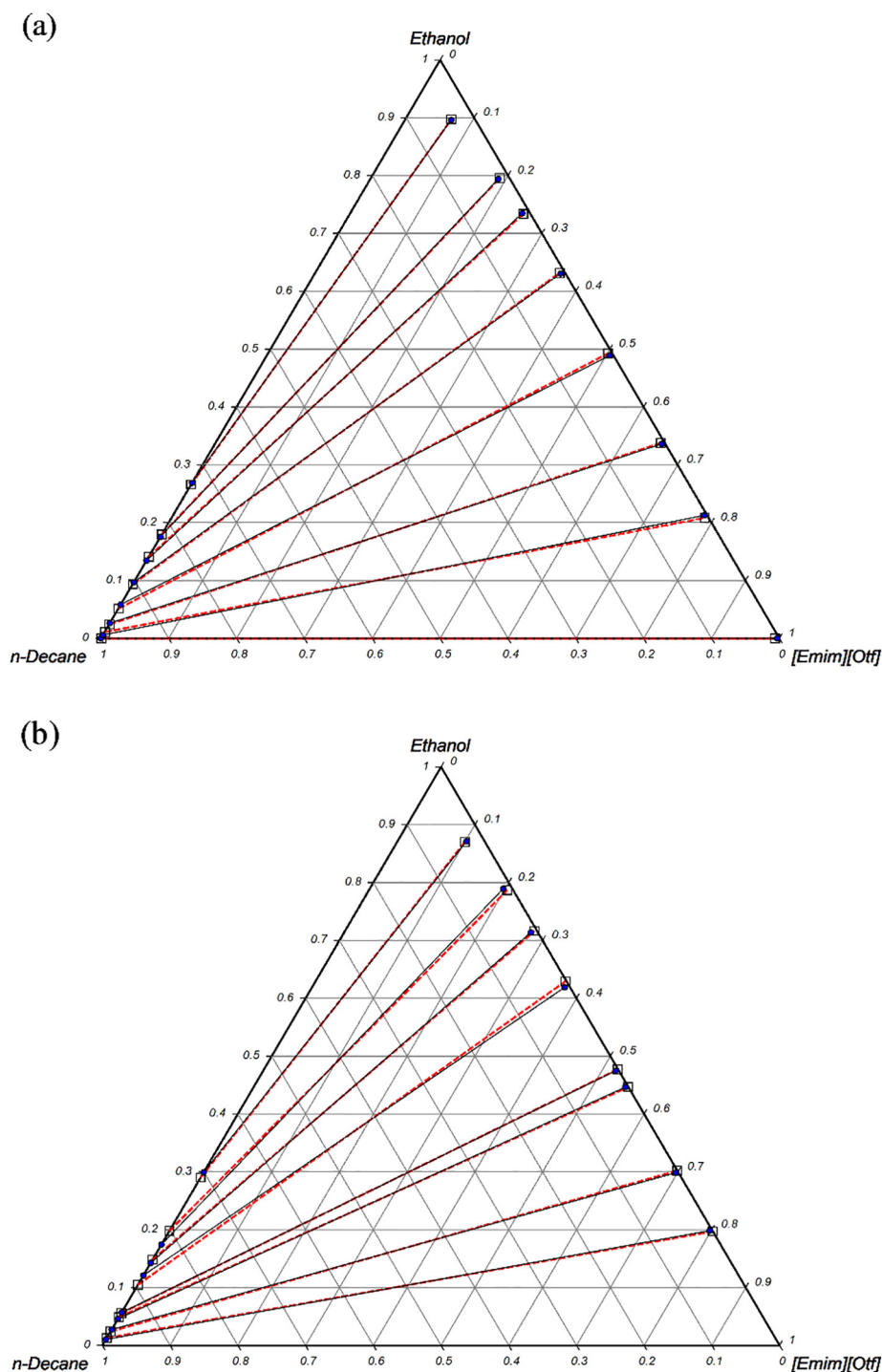


Fig. 2. Tie-lines of *n*-decane/ethanol/[Emim][Otf] ternary systems (a) at $T = 298.15$ K; (b) $T = 313.15$ K, and $p = 101.3$ kPa. (— with circles: experimental data; — with squares: NRTL model.)

To study the feasibility of using ILs as solvent in liquid-liquid extraction process, two parameters of solute distribution ratio (β) and selectivity (S) were evaluated. The solute distribution ratio indicates the capacity and required amount of solvent to separate ethanol from its mixture with *n*-decane. Also, the selectivity is a criterion to show power of solvent for extracting process. The number of equilibrium stages are needed to extract ethanol from *n*-decane are decreased when the selectivity increases. The β and S are calculated as follows:

$$\beta = \frac{x_2^{II}}{x_2^I} \quad (1)$$

and

$$S = \frac{x_1^I x_2^{II}}{x_1^{II} x_2^I} \quad (2)$$

where x_1 and x_2 represent the mole fraction of *n*-decane and ethanol, respectively. The superscripts I and II refer to *n*-decane-rich (upper phase) and IL-rich (lower phase) phases, respectively. The calculated values of β and S are tabulated in Table 2. Also, the obtained values of β and S are illustrated versus ethanol mole fraction of upper phase in Figs. 3 and 4, respectively.

According to Table 2, all values calculated for S are much more than unity, implies the possibility of using [Emim][SCN] and [Emim][Otf] for ethanol extraction from upper phase. High values are expected for both parameters to consider an IL as an ideal solvent for extraction process. As shown in Figs. 3 and 4, higher values are obtained for solute distribution ratio and selectivity at low ethanol mole fractions. Which means, the easier separation of ethanol from its mixtures with *n*-decane. Moreover, the β and S values decreases, as the mole fraction of ethanol increase in *n*-decane-rich phase. As can be seen in Fig. 3, higher values of solute distribution ratio are obtained for the *n*-decane/ethanol/[Emim][Otf] ternary system in comparison to *n*-decane/ethanol/[Emim][SCN] at the both studied temperatures. Moreover, as presented in this figure, β values of investigated ternary systems are slightly decreased as the temperature increased. As illustrated in Fig. 4, the IL with [SCN] anion is more selective than IL with [Otf] anion. Furthermore, the represented results in Fig. 4 show that increasing temperature from 298.15 K to 313.15 K, has no considerable effect

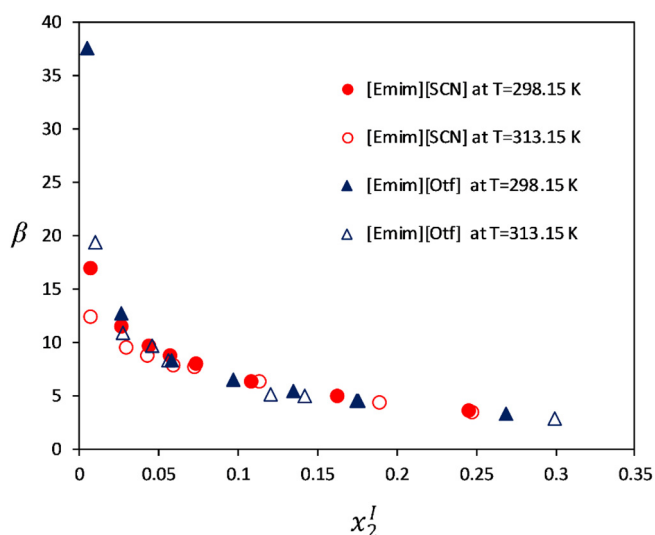


Fig. 3. Solute distribution ratio for the ternary systems: *n*-decane/ethanol/[Emim][SCN] (●) at $T = 298.15$ K, (○) at $T = 313.15$ K, *n*-decane/ethanol/[Emim][Otf] (▲) at $T = 298.15$ K, and (△) at $T = 313.15$ K, as a function of ethanol mole fraction in *n*-decane-rich phase.

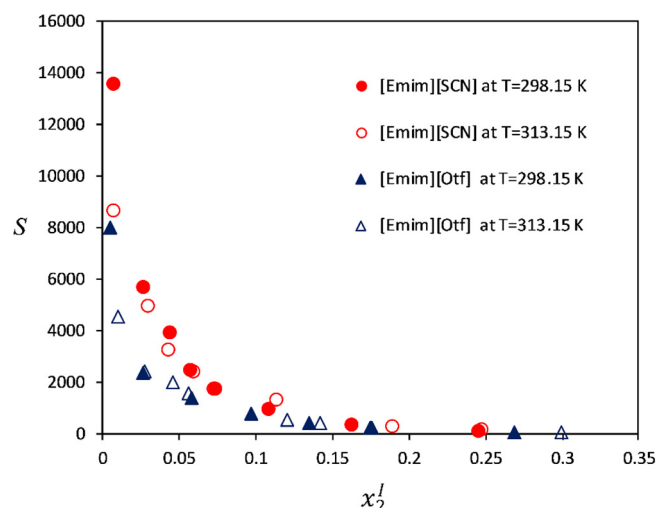


Fig. 4. Selectivity for the ternary systems: *n*-decane/ethanol/[Emim][SCN] (●) at $T = 298.15$ K, (○) at $T = 313.15$ K, *n*-decane/ethanol/[Emim][Otf] (▲) at $T = 298.15$ K, and (△) at $T = 313.15$ K, as a function of ethanol mole fraction in *n*-decane-rich phase.

on selectivity of solvents. The high values of β and S reported in this work, indicate both ILs ([Emim][SCN] and [Emim][Otf]) are effective to extract ethanol from ethanol-*n*-decane mixtures. Although, utilizing [Emim][Otf] as extractive agent needs less amount of solvent to separate ethanol from its mixture with *n*-decane. However, using [Emim][SCN], results in higher purity product after solvent regeneration.

3.2. Thermodynamic correlation

Non-random two liquid (NRTL) model has been successfully applied to predict the phase behavior of systems containing ILs [29]. The activity coefficient is used to consider the non-ideality of components exist in liquid phases. The equilibria between *n*-decane-rich and IL-rich phases can be considered as follows:

$$x_i^I \gamma_i^I = x_i^{II} \gamma_i^{II} \quad (3)$$

where x_i and γ_i respectively, are mole fraction and activity coefficient of component i ; and superscripts I and II refer to upper phase and lower phase, respectively. In this work, activity coefficients are calculated at known compositions and temperature as follows by utilizing NRTL model:

$$\ln \gamma_i = \left(\frac{\sum_{j=1}^n \tau_{ji} x_j G_{ji}}{\sum_{k=1}^n x_k G_{ki}} \right) + \sum_{j=1}^n \left(\frac{x_j G_{ij}}{\sum_{k=1}^n x_k G_{kj}} \right) \left(\tau_{ij} - \frac{\sum_{m=1}^n \tau_{mj} x_m G_{mj}}{\sum_{k=1}^n x_k G_{kj}} \right) \quad (4)$$

where

$$G_{ij} = \exp(-\alpha_{ij} \tau_{ij}) \quad (5)$$

and

$$\tau_{ij} = \frac{g_{ij} - g_{ii}}{RT} = \frac{\Delta g_{ij}}{RT} \quad (6)$$

where α_{ij} is the non-randomness parameter, Δg_{ij} is the binary interaction parameter, R is the gas constant and T is the absolute temperature. The adjustable parameters of NRTL model (α_{ij} , Δg_{ij} and Δg_{ji}) were optimized for the investigated system in this work through the minimizing the difference between LLE data and mole fractions calculated from model, known as objective function (OF), which is defined as follows:

Table 3The binary interaction parameters of NRTL model and deviations for LLE data of the ternary systems at $T = 298.15$ – 313.15 K and pressure of 101.3 kPa.

T/K	i - j	$\Delta g_{ij}/\text{kJ}\cdot\text{mol}^{-1}$	$\Delta g_{ji}/\text{kJ}\cdot\text{mol}^{-1}$	α_{ij}	σ_x
n -Decane (1) + Ethanol (2) + [Emim][SCN] (3)					
298.15	1-2	-6.567	13.642	0.040	0.35
	1-3	14.217	9.863		
	2-3	6.285	-10.410		
313.15	1-2	0.374	7.814	0.262	0.59
	1-3	10.152	17.777		
	2-3	7.367	-7.906		
n -Decane (1) + Ethanol (2) + [Emim][Otf] (3)					
298.15	1-2	-2.970	11.751	0.115	0.28
	1-3	12.935	5.586		
	2-3	7.161	-9.720		
313.15	1-2	-1.600	7.899	0.085	0.69
	1-3	46.291	17.345		
	2-3	8.294	-10.000		

$$OF = \sum_{i=1}^M \left[\left(x_{1i}^{I,calc} - x_{1i}^{I,exp} \right)^2 \right] + \left[\left(x_{2i}^{I,calc} - x_{2i}^{I,exp} \right)^2 \right] + \sum_{i=1}^M \left[\left(x_{1i}^{II,calc} - x_{1i}^{II,exp} \right)^2 \right] + \left[\left(x_{2i}^{II,calc} - x_{2i}^{II,exp} \right)^2 \right] \quad (7)$$

where x_1 and x_2 are *n*-decane and ethanol mole fractions, respectively; and M is the number of experimental tie-lines. The value of root mean square deviation (σ_x), defined below, shows the capability of using NRTL model to correlate the experimental equilibria data.

$$\sigma_x = 100 \left(\frac{\sum_{i=1}^M \sum_{j=1}^{N-1} \left(\left(x_{ij}^{I,calc} - x_{ij}^{I,exp} \right)^2 + \left(x_{ij}^{II,calc} - x_{ij}^{II,exp} \right)^2 \right)}{2MN} \right)^{1/2} \quad (8)$$

Here N equals the number of components present in the mixture. The obtained adjustable parameters of NRTL model with σ_x are reported in Table 3.

Moreover, the calculated data from NRTL model are compared with experimental data in Figs. 1 and 2. The low reported value of σ_x shows the NRTL model has fairly correlated the experimental data of *n*-decane/ethanol/[Emim][SCN] and *n*-decane/ethanol/[Emim][Otf] systems studied in this work.

4. Conclusion

In this work, the experimental LLE data were recorded for four ternary systems: *n*-decane/ethanol/[Emim][SCN] and *n*-decane/ethanol/[Emim][Otf] at two temperatures ($T = 298.15$ K and $T = 313.15$ K) and the pressure of 101.3 kPa. The effectiveness of [Emim][SCN] and [Emim][Otf] as extractive agents for separation of *n*-decane-ethanol mixtures were investigated through the values of solute distribution ratio (β) and selectivity (S). The high values of S and β indicate the potential of [Emim][SCN] and [Emim][Otf] as solvents to extract ethanol from its mixture with *n*-decane. The effect of temperature on extraction parameters (β and S) were also investigated. It was found that solute distribution ratio decreased slightly by increasing temperature. However, no significant changes observed in selectivity values of the investigated ILs in this work, as the temperature increased. Using [Emim][Otf] leads to higher values of β relative to [Emim][SCN]. Although, higher values of selectivity are reported for [Emim][SCN] in comparison to [Emim][Otf]. Finally, experimental tie-lines have been correlated by NRTL model with sufficient accuracy.

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